

Discussion

1. Generalizability and Clinical Translation

Applicability Across Monogenic Classes

De-Risking Mechanism

Timeline Compression

Beneficiaries

2. Limitations

Surrogate Database Size

Structural Model Confidence

Immunogenicity Prediction Accuracy

Patient-Specific Immunogenicity

Codon Optimization and Translational Effects

GTEx Bulk Tissue Limitations

3. Regulatory, CMC, and IP Implications

Regulatory Pathway Acceleration

CMC Implications

Intellectual Property and Freedom-to-Operate

Cost and Platform Economics

4. Future Directions

CRISPR Base Editing and Prime Editing

Non-Viral Delivery Platforms

Machine Learning Refinement

Patient-Specific Immunogenicity

Regulatory Integration

Long-Term Vision